
The Tangent and Velocity Problems

Work these in class. Worked solutions are on the posted slide deck — each problem below names the slide to look for.

2.1 • Problem 1*slide: First: What the Parentheses Mean in $f(x)$*

With $f(x) = x^2 + 1$, write $f(5)$, then $f(a)$, then $f(x + h)$.

2.1 • Problem 2*slide: First: What the Parentheses Mean in $f(x)$*

Same function or different? $g(x) = 3x - 4$ and $g(s) = 3s - 4$.

2.1 • Problem 3 (worked together)*slide: First, Lines*

Through $(1, 2)$ with slope 3: point-slope is $y - 2 = 3(x - 1)$, and expanding gives $y = 3x - 1$.

2.1 • Problem 4*slide: First, Lines*

Find the line through $(3, 9)$ with slope 6 — once each way. Do you get the same line?

2.1 • Problem 5*slide: What Is a Tangent Line?*

Euclid's definition — *a line meeting the curve exactly once* — is fine for the circle. Use the right-hand picture to break it. There are two separate things wrong.

2.1 • Problem 6 (worked together)*slide: Sliding Q Toward P*

The top row, so the pattern is clear: at $x = 6$, $m_{PQ} = \frac{6^2 - 9}{6 - 3} = \frac{27}{3} = 9$.

2.1 • Problem 7

slide: Sliding Q Toward P

Two rows of each table are done. Compute $m_{PQ} = \frac{x^2 - 9}{x - 3}$ for $x = 3.5, 3.1, 3.01$ and for $x = 2.9, 2.99$ — then answer the question that matters: what are the slopes *doing*?

2.1 • Problem 8 (worked together)

slide: Evaluating the Limit

Why we are allowed to cancel $(x - 3)$ is the whole point, so let me do this one.

2.1 • Problem 9

slide: Average Velocity over Shrinking Intervals

Do not compute another one. Say what number these are heading for, and say how confident you are.

2.1 • Problem 10

slide: They Are the Same Problem

In the picture, one line's slope is the *average* velocity over $[a, a + h]$ and the other's is the *instantaneous* velocity at $t = a$. Which is which? Then say what happens to the secant as h shrinks toward 0.